Question 1 [7 Marks]:
Simplify the following expression using **tabulation** method ONLY: F(A,B,C) = A'B'C' + A'B'C + AB'C + ABC' + ABC

Question 2 [8 Marks]:
Design a circuit diagram for the following comparator system that takes three 3-bit binary numbers [A, B and C] as inputs and outputs in the following fashion:
Output=1; if A-5>2B
Output=0; otherwise
NB1: You need to show a gate level diagram for the comparator, meaning you cannot use block diagram of magnitude comparator. However, you may use block diagrams of any other ICs you require.
NB2: Drawing the circuit diagram horizontally in your script may help you.
NB3: You cannot change the equation.You must draw the circuit for the given equation.

Question 3 [5 Marks]:
Construct the circuit of the following function using 3:8 decoder and 2:4 decoder only:
F(A,B,C,D) =(A+B')(C'+D')
NB: You must use both the decoders in a single circuit. Your circuit should be cost efficient, meaning you have to use the lowest number of components possible. You should use external gates for connecting inputs/outputs if required.

Question 4 [10 Marks]:
Consider the following sequential circuit. Determine its state table and state diagram.